

Circuits and Electronic Laboratory

Experiment #2

Purpose of Experiment

Gaining the habit of building circuits on the breadboard, learning the relation between ammeter and current on the circuit and experimenting Kirchhoff's current law.

General Information

Kirchhoff's Current Law: This law states that on any circuit at any time t sum of currents that originates from the node and currents that goes into the node is zero.

$$\sum_{k=0}^n I_k = 0 \quad (1)$$

where n is the total number of branches with currents flowing towards or away from the node. Currents that goes into the node have positive sign. Vice versa currents that goes out of the node has negative sign.

A device called ammeter (**Ampere Meter**) is a measuring instrument used to measure the current in a circuit. Electric currents are measured in amperes. Today there are two different kinds of ammeter which are analog and digital ones. Digital ones are widely used nowadays. An ideal ammeter has an internal resistance of zero. Due to this property it behaves like a short circuit. Practical ammeters usually has 0.01Ω - 0.1Ω Ammeters always connected in series to the component that needs its current to be measured.

Do not connect ammeter parallel to component. Be sure about the current to be measured is smaller than ammeters maximum current. If not sure, first use biggest current ratio of ammeter to measure.

Nodes: a node in a circuit is defined as a region between two circuit elements. An example of 3 different nodes in a circuit given Figure 1. Each color represents different node.

Tip: If a components current needs to be measured, follow this steps:

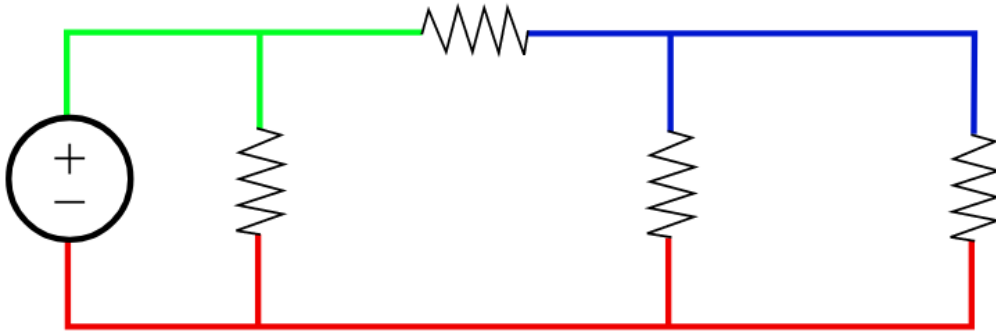


Figure 1: Nodes In a Circuit, source: Wikipedia

- Get the multimeter into the current measurement state.
- Put red (+) probe to mA input if current to be measured is smaller than 200ma. If it's bigger than that, or current is unknown use A input of multi meter. Put black (-) probe to comm (gnd) input.
- Turn off all voltage sources.
- Disconnect the cable that connects the component.
- Connect ammeter appropriately instead of the cable that disconnected.
- Turn on voltage sources. Read the current value.

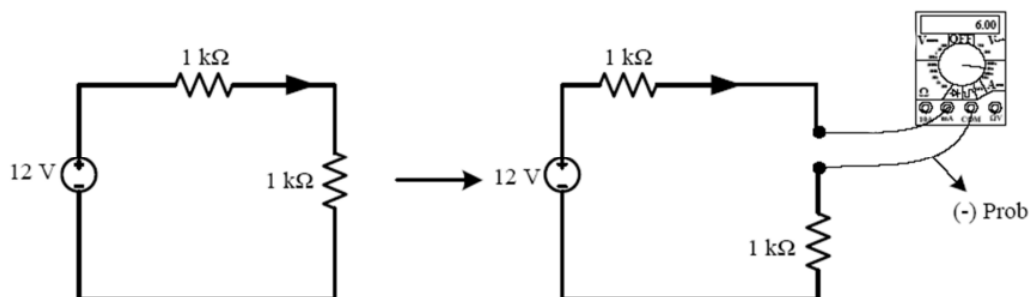


Figure 2: Sample connection diagram of a ammeter

Preparations Before Experiment

- For the circuit in Figure 3, calculate currents in Table 1 with given voltages.
- Perform simulation studies for Figure 3 and Figure 4 to understand the experiment, and check your calculated values with the findings of the simulations.
- According to all tables in the lab sheet, take note of the numerical results of the simulations on your lab sheet even if there is not a suitable table to fill.

Section 1

1. Construct the circuit given in Figure 3. Set the DC source with the help of voltmeter according to Table 1. Measure I_k , I_1 , I_2 , I_3 with the help of ammeter. Fill the Table 1.

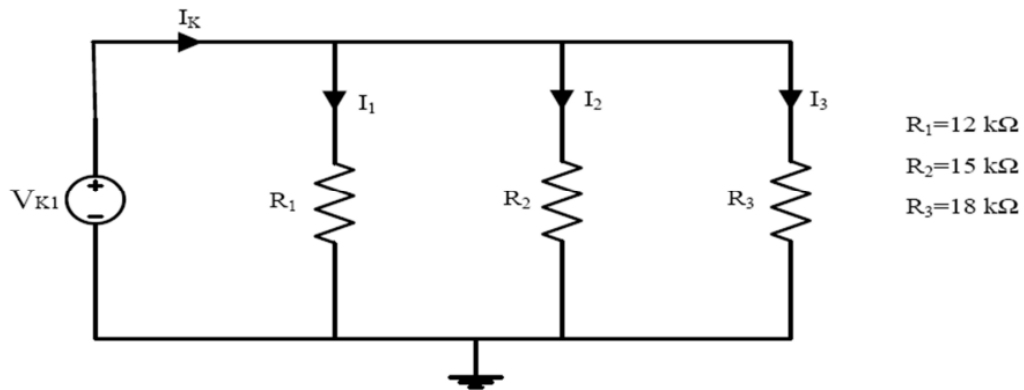


Figure 3:

V_{k1}	$I_1[A]$	$I_2[A]$	$I_3[A]$	$I_k[A]$ (Mea- sured)	$I_k[A]$ (Cal- culated)	Relative Error [%]
5V						
8V						
12V						
Table 1						

2. In the circuit depicted in Figure 4, set source voltage to 12V. Fill the Table 2 by measuring all component currents. Show that Kirchhoff's current law holds.

$I_{R1}[A]$	$I_{R2}[A]$	$I_{R1} + I_{R2}[A]$	$I_{R3}[A]$	$I_{R4}[A]$	$I_{R3} + I_{R4}[A]$	$I_k[A]$
Table 2						

2. Measure the voltages of all components in Figure 4, and fill the Table 3.

$V_{R1}[V]$	$V_{R2}[V]$	$V_{R3}[V]$	$V_{R4}[V]$
Table 3			

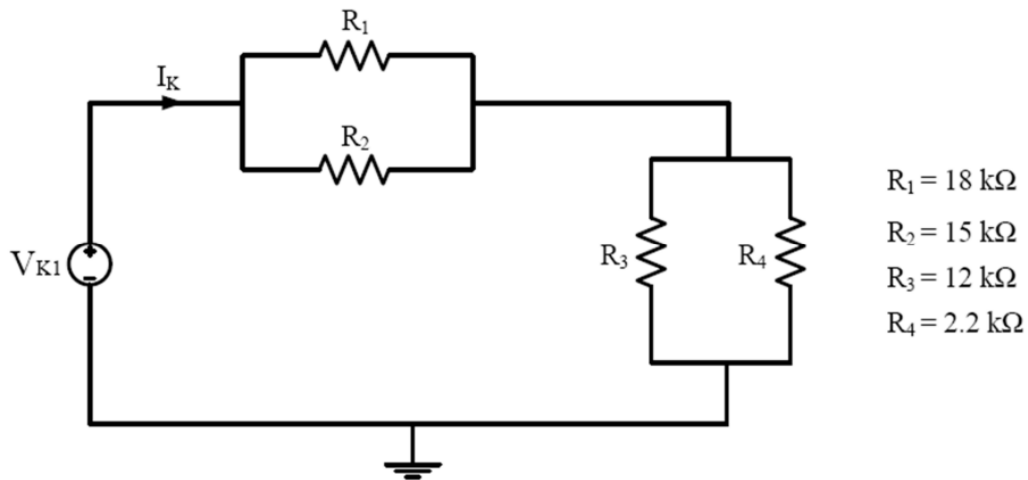


Figure 4:

Section 2

1. Find the three nodes on the circuit in Figure 4, and fill Table 3 with Kirchhoff's current law equations for the nodes. Annotate nodes starting from voltage source with left to right. Ex: Node 1 is the node between voltage source and R_1 , R_2 .

	Symbolic Equation	Numerical Equation
Node 1		
Node 2		
Node 3		
Table 4		

2. Find the three cycles of the circuit in Figure 4, and fill the Table 5 with Kirchhoff's voltage law equations.

	Symbolic Equation	Numerical Equation
Cycle 1		
Cycle 2		
Cycle 3		
Table 5		